

SNAPDRAGON AI LAB BUILD & PRESENT CHALLENGE

A Proposal to Qualcomm & HP

The Enterprise Data Sentinel

An On-Device, Zero-Trust DLP Copilot for
Snapdragon-Powered HP PCs

Category: Enterprise Security & Productivity — Edge AI

Target Platform: Snapdragon X Elite / X Plus / X2 Plus — Hexagon NPU (45–80 TOPS)

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1. Executive Summary

Current Data Loss Prevention (DLP) tools face a fundamental catch-22: they either fail to act in real time, or they solve the security problem by uploading the very sensitive data they are supposed to protect to a third party. We are introducing "Enterprise Data Sentinel," a solution designed to break this cycle. By using a three-tier AI inference cascade—comprising OCR, Named Entity Recognition, and a lightweight contextual reasoning language model—the system runs entirely on the local Hexagon NPU. This allows the software to intercept clipboard, screen share, and file egress events before they are completed, redacting or blocking sensitive information at the source without any data ever leaving the device. Built on Windows ML using the Qualcomm QNN Execution Provider and models validated by the Qualcomm AI Hub (EasyOCR, DistilBERT, and Qwen3-0.6B via GenieX), this is exactly the type of continuous, real-time, privacy-first AI inference that on-device NPUs were built for—something a cloud solution simply cannot replicate. The result isn't just a faster DLP tool; it's an entirely new category of product that repositions Snapdragon-powered HP PCs as the premier secure workstation for today's compliance-heavy enterprise..

2. Problem Statement & Enterprise Need

2.1 Data Leakage Is a Silent, Systemic Enterprise Risk

Data exposure driven by insiders not by hackers is a cause of big company breaches. These Data exposures rarely show up as dramatic thefts; instead they happen through normal daily mistakes. It can be a support agent pasting a social security number into a third-party chatbot to say "summarize a ticket " an engineer sharing a screen that shows a live API key or a finance analyst uploading an unredacted payroll list to a private cloud drive to work from home. Because these actions are small and normal they are often unseen, by border and network security tools.

2.2 Cloud DLP Is Architecturally Self-Defeating

The standard DLP model today works by routing suspicious content—like clipboard text, screenshots, or uploaded files—to a cloud-based classification API for inspection. This creates an inherent paradox that no amount of cloud infrastructure can fix:

- * Data residency and sovereignty risks: Sending content to a cloud endpoint for scanning is, in itself, a data transfer. Under regulations like the EU GDPR or India's Digital Personal Data Protection (DPDP) Act, this transfer—even if it is temporary or intended for "security" purposes—can trigger a compliance violation if the destination, sub-processor, or retention policies aren't perfect. The tool meant to reduce liability becomes a new source of it.

- * Latency prevents real-time prevention: A clipboard paste or a file upload happens in milliseconds. A round trip to the cloud for classification—which usually takes anywhere from 200ms to several seconds depending on the network—is far too slow to stop the action before it finishes. As a result, most cloud DLP is reactive: it logs and alerts you to a leak that has already happened, rather than actually preventing it.

- * Scaling issues with bandwidth and cost: Constantly monitoring every clipboard event, screen share, and file transfer across an entire enterprise requires massive outbound bandwidth. If every event is streamed to the cloud for inference, the costs multiply linearly with every new employee, making comprehensive, always-on coverage financially impossible for most companies.

* Unreliable availability: High-security environments—such as defense contractors, financial trading floors, healthcare facilities, or aircraft in flight—often operate with restricted or zero internet connectivity. A DLP tool that relies on the cloud effectively shuts itself off in the exact environments where security is most critical.

2.3 Why This Is a Fundamental (Not Incidental) Case for Edge AI

This isn't a workload that just happens to work better locally; it is a workload where cloud execution is a logical contradiction. The entire goal of a DLP system is to prevent sensitive content from leaving a controlled environment. A cloud-based system must first move that content across the boundary to do its job. Only an architecture where classification and decision-making happen instantaneously on the same silicon that holds the data—with zero network dependency—can provide DLP that is truly real-time, preventative, and private by design. This is the exact gap that the Hexagon NPU in Snapdragon-powered HP PCs is designed to close.

3. Proposed Solution Architecture

The Enterprise Data Sentinel follows a key rule. Sensitive content must never leave the device before it is classified. The classification must happen before any risky action is finished. The Enterprise Data Sentinel uses an event-driven pipeline. Nothing runs constantly in the background except a very low-power listener. Each inference stage, in the Enterprise Data Sentinel is triggered when a particular user action calls for it.

3.1 Workflow

Step	Trigger / Component	Action Taken
1	System event hook (Win32 Add Clipboard Format Listener; screen-share start via foreground/session APIs; shell-level file-egress hook)	Fires the instant a user copies, cuts, initiates a screen share, or begins a file upload/attach action.
2	In-memory content capture	The clipboard payload or a frame of the shared/uploaded content is staged in volatile memory only — never written to disk — for inference.
3	Tier 1 — OCR (conditional)	If the payload is image-based (screenshot, PDF snippet, photographed document), easyocr-small-stage1 (INT8) extracts text regions. Skipped entirely for native text — event-driven, not always-on.
4	Tier 2 — NER / Classification	Extracted or native text is passed to DistilBERT-Base-Uncased-HF, fine-tuned to tag PII, PHI, financial identifiers, and credential/secret patterns with per-token confidence scores.
5	Tier 3 — Contextual Adjudication (conditional)	Only for entities Tier 2 flags as ambiguous, Qwen3-0.6B (w4a16, via GENIE) reasons over surrounding context — sentence, document type, destination app — to render a final verdict and a plain-language justification.
6	Policy engine decision	Sensitivity verdict is checked against locally stored, IT-configured policy (thresholds, allow/deny lists, destination-based rules).
7	Enforcement (WinUI 3 layer)	Allow silently , Redact/mask sensitive spans in place • Block the action with an on-screen policy citation • Log the event locally (encrypted, IT-configurable retention) for audit.

3.2 Design Principles

- * Event-driven not polling-based:** Event-driven design keeps power usage low by staying idle until a system event happens such as writing to the clipboard starting a screen share or opening a file dialog.
- * In-memory In-memory only storage:** means that raw captured content is never written to disk, which removes the risk of forensic leaks or leftover data.
- * Progressive escalation:** Progressive escalation means that each level is only used if the level before it does not have confidence to make a clear decision. See Section 4 for details.
- * Policy, local enforcement:** Local policy local enforcement ensures that all decisions—rules, thresholds and lists—are made directly on the device using a locally cached policy file. Only anonymous configuration data, never user content syncs from MDM consoles such as Intune and it only happens on a schedule set by an administrator.

4. Technical Implementation & Qualcomm AI Hub Integration

The Enterprise Data Sentinel uses a three-tier inference cascade designed to invoke models only when content ambiguity requires it. This approach is intentional: it makes an "always-on" DLP process thermally and economically viable on a laptop, avoiding the need for data-center-grade hardware.

4.1 The Three-Tier Cascade

Tier	Model	Size / Precision	Runtime	Invocation
1	EasyOCR (easyocr-small-stage1)	INT8 quantized	Windows ML + QNN EP	Image-bearing content only
2	DistilBERT-Base-Uncased-HF	11.3M params, ~11 MB, INT8	Windows ML + QNN EP	Every text payload
3	Qwen3-0.6B	w4a16, ~23.9 tok/s	GenieX (GENIE runtime)	Ambiguous entities only (est. <10% of events)

Tier 1 — Text Extraction (EasyOCR, easyocr-small-stage1, INT8)

This tier is only activated when the payload is image-based, such as a screenshot, a shared application window, or a photo of a document. Text-native clipboard events skip this step entirely, ensuring that no OCR compute is wasted on standard copy-paste actions.

Tier 2 — Token Classification (DistilBERT-Base-Uncased-HF, 11.3M params, ~11 MB INT8)

This is the system's workhorse. This compact transformer is fine-tuned to recognize PII (names, IDs), PHI (medical records), financial data (card numbers), and secrets (API keys). It processes the vast majority of content in milliseconds. Its tiny ~11 MB footprint allows it to stay resident in NPU-adjacent memory without impacting power or heat.

Tier 3 — Contextual Adjudication (Qwen3-0.6B, w4a16, via GENIE)

This is reserved for the "long tail" of difficult cases—like a nine-digit number that could be either a Social Security Number or a random ticket ID, or a code snippet that might be a real secret or just a placeholder. Only here does the system use a generative, reasoning-capable model to analyze context and provide an explainable verdict. Since Tier 2 handles most cases, Tier 3 is estimated to run for less than 10% of total events in a typical environment.

4.2 Why a Cascade Outperforms a Monolithic LLM

*Why a Cascade Outperforms a LLM

A simpler design might route every event through one massive LLM. The Enterprise Data Sentinel rejects this approach for three reasons:

- * Compute proportional to difficulty: Most content is either clearly harmless like a URL or a meeting time or clearly sensitive like a credit card number. Neither needs a reasoning model. Tier 2 can resolve them in single-digit milliseconds. Using an LLM for everything is a waste of resources with no gain in accuracy.

- * Power and thermal management: Running a LLM on every keystroke or screen change would keep the NPU or CPU at high utilization all day. This would drain the battery. Cause thermal throttling. The cascade ensures the heaviest models stay dormant until they are actually needed.

- * Real-time latency: DLP must decide before a paste or upload is completed. Tier 2's millisecond-scale speed keeps the user experience seamless for 90% or more of events. By reserving the generative tier for rare flagged cases any added latency remains nearly imperceptible, to the user.

Ultimately the cascade isn't a compromise. It is an architecture. It is only possible because each tier can be independently and efficiently managed by the NPU.

4.3 Software Stack

- * **C# / WinUI 3: Manages the user interface and shell, handling event hook registration, the redaction/blocking UI, policy configuration, and the local audit-log viewer.**

*** Windows ML + Qualcomm QNN Execution Provider: Tiers 1 and 2 are deployed as ONNX models. The QNN Execution Provider is automatically registered through the ONNX Runtime, allowing inference graphs to be routed to the Hexagon NPU seamlessly without the need for complex, low-level integration.**

*** GenieX (GENIE runtime): Hosts the Tier 3 generative model. It is specifically designed for running small language models on-device using w4a16 quantization, offering a streamlined prompt-in/verdict-out interface for the C# orchestration layer.**

4.4 Role of Qualcomm AI Hub

Every model in the processing cascade is sourced and validated through the Qualcomm AI Hub. Each model is pre-optimized—meaning it has been quantized, layer-fused, and profiled—specifically for the Hexagon NPU. We use published on-device latency and accuracy benchmarks to choose the perfect model variant; for instance, selecting easyocr-small-stage1 over larger OCR models to ensure we meet the sub-perceptible latency required for this application. This approach eliminates the biggest risk in edge-AI development: the uncertainty of whether a model will actually meet real-time performance targets on the actual silicon before it is deployed.

5. Snapdragon Hardware Acceleration

The three-tier cascade described in Section 4 isn't just compatible with the Hexagon NPU—it is only viable as a continuous enterprise product because of the specific architecture of Snapdragon's NPU. This section outlines that hardware dependency.

5.1 Micro-Tile Inferencing: Eliminating DRAM Thrashing

The Hexagon NPU performs micro-tile inferencing by combining network layers into a single tile that fits completely inside dedicated on-chip shared memory. Because of this intermediate activations for a fused tile never need to move forth to system DRAM. This is a benefit for a background workload that stays in memory. In a CPU or discrete GPU pipeline each layer's activations must shuttle in and out of shared system memory fighting for bandwidth with everything the user is doing—such as video calls, browser tabs or IDE builds. Running a three-model cascade in that way would clearly slow down foreground apps. Micro-tile inferencing keeps Sentinel's compute footprint invisible to the rest of the system, which is crucial, for a security tool that must run continuously without users choosing to turn it off for performance reasons.

5.2 INT8 and w4a16 Quantization: Footprint and Throughput

Tiers 1 and 2 run on INT8 models while Tier 3 uses w4a16 (weights and sixteen-bit activations) through GenieX. Compared to FP16 or FP32 execution this cuts both the memory footprint and the computational cost of each model by four times or more. In terms this lets all three tiers—OCR, NER and a 0.6-billion-parameter reasoning model—remain in memory and ready for scheduling, without requiring dedicated VRAM or pushing other applications' memory. This level of efficiency is simply impossible when a design relies on a GPU.

5.3 Event-Driven Duty Cycling via the Sensing Hub

Qualcomm's Sensing Hub uses a micro-NPU to keep the system-event listener running on an ultra-low-power, always-on core. This core is completely separate from the Hexagon NPU. The system-event listener checks for things, like clipboard writes, screen-sharing or file egress. It operates independently. Doesn't need the main NPU to be active all the time. The main NPU only turns on when a real event happens. This design lets the Sentinel stay "on" without using too much power. There's no idle power draw because nothing is constantly checking, sampling or making guesses in the background. It stays quiet and efficient until something actually needs attention.

5.4 Battery Efficiency vs. CPU Thermal Throttling

A CPU-based version of this cascade would create a direct conflict between security and usability. Constant inference on CPU cores drives up die temperature, which triggers thermal throttling. This reduces clock speeds and degrades the performance of the applications the user is actually trying to work in. Over a full workday, this would also significantly drain battery life, creating a predictable failure mode where users or IT departments disable the protection just to maintain performance. NPU offloading avoids this entirely. The inference cascade runs within a power envelope designed for sustained, all-day operation. This is the core advantage of Snapdragon: the NPU doesn't just make the Sentinel faster; it makes an always-on version of the Sentinel possible without interfering with the user's workday..

5.5 Platform Prerequisites

Requirement	Specification
Silicon	Snapdragon X Elite / X Plus / X2 Plus — Hexagon NPU, 45–80 TOPS
Operating System	Windows 11 24H2 (build 26100 or later)
NPU Driver	Hexagon NPU driver 30.0.140.0 or later
Always-on sensing	Qualcomm Sensing Hub (dual micro-NPU) for event listening

6. Deployment, Scalability & Security

6.1 Deployment Model

The Enterprise Data Sentinel comes as one installer (MSIX/MSI). It works for installs and for many computers via tools like Microsoft Intune or VMware Workspace ONE. All three levels are pre-quantized so the whole model weighs, than 100 MB. That keeps the installation size. Updates very small. Most important there is no cloud back-end, no API key management and no per-seat cloud subscription needed for the DLP functions. This eliminates the friction points and ongoing costs that happen with ordinary DLP rollouts.

6.2 Zero Data Egress Guarantee

The core architectural claim—and the solution's biggest differentiator—is that no user content, in any form (whether raw, redacted, hashed, or embedded), is ever transmitted off the device for DLP functions. The only network activity the application generates is periodic, IT-scheduled synchronization of policy configurations, such as thresholds, allow/deny lists, and destination rules. By design, this data contains no user content. This isn't a promise that relies on blind trust; it is a concrete, auditable claim that enterprise security teams and hackathon judges can independently verify through standard network traffic inspection.

6.3 Offline Resilience

Since every inference tier runs locally, the Sentinel remains fully functional even with zero network connectivity. This isn't just a fallback mode; it is the primary way the system operates. This makes the solution ideal for the exact environments where data protection is most critical: air-gapped government and defense networks, financial trading floors with restricted connectivity, healthcare facilities, aircraft, and remote industrial or field sites.

6.4 Scalability & Total Cost of Ownership

Because all the thinking work is done by the endpoints chip the solution grows smoothly without any extra costs in the cloud as a company gets bigger. This is very different from cloud DLP ways, which often charge for each check or each person making costs go up as more people join. This is the part of how the solution can grow: the same setup, the same program and the same way of keeping things safe work the same for a small team of ten people and for a

big company with a hundred thousand workers, without needing to add more cloud systems or worry about managing them.

6.5 Security of the Tool Itself

- * Local, encrypted audit logs with IT-configurable retention—these are never uploaded and can only be viewed via the local console or deliberately exported by an administrator.**
- * Signed, integrity-checked model weights to prevent any tampering with the classification models.**
- * Least-privilege execution: the application only requests the specific OS permissions needed for clipboard and window-event hooking, requiring no elevated system privileges beyond what Windows' own accessibility and clipboard APIs demand.**

7. Conclusion & Impact

What ultimately differentiates all three pillars of this challenge from a competition is their enterprise-grade value proposition. For Technical Implementation, the final implementation provides a production-quality inference cascade, validated on the Qualcomm AI Hub, purpose-tuned for the Hexagon NPU's micro-tile architecture-not just a demo-on a laptop. For Application Use Case & Innovation, what we are solving for has universal relevance: every regulated industry and every remote workforce facing GDPR, DPDP, HIPAA and other data protection laws has the same exposure, and none of today's cloud AI tools can deliver it without exposing the very concepts of cloud and AI. And finally, on Deployment & Accessibility, the final implementation is delivered as one lightweight installer with no cloud backend, no per-seat inference bill, and no connectivity requirements-suitable for deployment in a small startup or a global enterprise fleet. As a demonstration of what on-device AI on Snapdragon enables at a system-wide level, I hope we can also say that this is the sort of thing we used to dream of when considering the next step in device evolution to put AI where it belongs: totally portable, totally autonomous. And if that's true, I think we can also say that the Snapdragon-powered HP PC is more than just an AI laptop-it is a truly secure enterprise workstation, where no data protection tool needs to upload data to protect it, where no tool needs to be turned off to save energy, and where work can truly be done productively with total peace of mind.

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